

Module 2: Transport Layer and Network Layer



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Transport Layer

- Services and Protocols
- UDP and TCP

Hands-on

- Sockets Introduction
- Elementary TCP Sockets
- TCP Client/Server Example
- I/O Multiplexing
 - select()
 - poll()
- Elementary UDP Sockets
- Advanced I/O Functions

Network Layer

- Introduction
- Network-layer Protocols
- Unicast Routing
- Multicast Routing
- Multicasting Basics
- Intra-domain Routing
- Inter-domain Routing
- Next Generation IP (IPv6)
- Quality of Service (QoS)

Hands-on

- Linux Routing Table
- Routing Caches
- Routing Cache Implementation
- Adding Routes using ip

Next Generation IP (IPv6)

- IPv4 address exhaustion created the need for a new Internet Protocol.
- In the early 1990s, IETF proposed:

Internet Protocol Version 6 (IPv6)

- Also called:
 - IPv6
 - IPng (IP Next Generation)

Why IPv6 Was Developed

Problems with IPv4

- Address depletion
- Large routing tables
- Limited Quality of Service (QoS)
- Weak built-in security
- Growing Internet users and devices

Major Improvements in IPv6

- ① Larger Address Space
- ② Better Header Format
- ③ New Options
- ④ Protocol Extensibility
- ⑤ Resource Allocation Support
- ⑥ Enhanced Security



Feature 1: Larger Address Space

IPv4 Address Length:

32 bits

IPv6 Address Length:



Number of Addresses:

$$2^{128}$$

Benefit

Provides virtually unlimited IP addresses for future Internet growth.

IPv4 vs IPv6 Address Space

Feature	IPv4	IPv6
Address Length	32 bits	128 bits
Number of Addresses	2^{32}	2^{128}
Address Format	Decimal	Hexadecimal

IPv6 provides a massive increase in address availability.

Feature 2: Better Header Format

- IPv4 stores options inside the header.
- Routers must process these options.
- Processing becomes slower.

IPv6:

- Separates options from the base header.
- Uses Extension Headers.
- Speeds up packet forwarding.



Feature 3: New Options

- IPv6 introduces new optional headers.
- Additional functionalities can be added easily.
- Routers process only required information.

Advantage

Improved flexibility and efficiency.

Feature 4: Protocol Extensibility

- IPv6 was designed for future expansion.
- New technologies can be supported through extension headers.
- No need to redesign the entire protocol.

Benefit

Supports future Internet applications and services.

Feature 5: Resource Allocation Support

IPv4 uses:

- Type of Service (ToS)

IPv6 replaces it with:

- Traffic Class
- Flow Label

These fields support:

- Voice Traffic
- Video Streaming
- Real-Time Applications



Traffic Class

- Specifies packet priority.
- Supports Quality of Service (QoS).

Flow Label

- Identifies packets belonging to the same flow.
- Useful for multimedia communication.

Feature 6: Enhanced Security

IPv6 provides support for:

- Authentication
- Data Integrity
- Encryption



Result

Secure and reliable communication over the Internet.

Why Has IPv6 Adoption Been Slow?

Three technologies delayed IPv4 exhaustion:

- ① CIDR (Classless Addressing)
- ② DHCP
- ③ NAT

These techniques extended the life of IPv4.

Need for IPv6 Today

Rapid growth of:

- Mobile Internet
- Cloud Computing
- IoT Devices
- IP Telephony
- Video Streaming

requires a much larger address space.



IPv6 is the long-term solution.

Applications Driving IPv6 Adoption

- Smart Homes
- Internet of Things (IoT)
- Mobile Networks
- Cloud Services
- 5G Communication
- Smart Cities



Benefits of IPv6

- Huge Address Space
- Simplified Header Processing
- Better Routing Efficiency
- Built-in Security Features
- Improved Quality of Service
- Better Mobility Support

IPv4 vs IPv6 Comparison

Feature	IPv4	IPv6
Address Size	32 bits	128 bits
Header Length	Variable	Fixed (40 bytes)
Security	Optional	Built-in Support
QoS Support	Limited	Improved
Configuration	Manual/DHCP	Auto Configuration
Address Space	Limited	Extremely Large

Topics Covered in IPv6

In this section we study:

- ① IPv6 Packet Format
- ② IPv6 Addressing
- ③ IPv4 to IPv6 Transition Mechanisms
- ④ ICMPv6

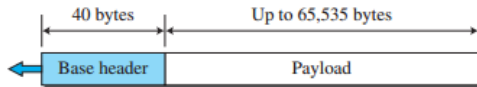
Summary

- IPv6 was introduced to overcome IPv4 limitations.
- It provides 128-bit addressing.
- Supports extensibility through extension headers.
- Introduces Traffic Class and Flow Label.
- Enhances security and QoS support.
- Forms the foundation of the future Internet.

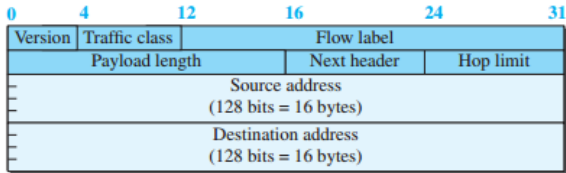
IPv6 Datagram Structure

- IPv6 packet consists of:
 - ① Base Header (Fixed 40 Bytes)
 - ② Payload
- Base Header contains essential routing information.
- Payload contains:
 - Extension Headers (Optional)
 - Upper Layer Data (TCP, UDP, ICMPv6)

Figure 4.101 *IPv6 datagram*



a. IPv6 packet



b. Base header

IPv6 Packet Structure

Version 4 bits	Traffic Class 8 bits	Flow Label 20 bits
Payload Length 16 bits	Next Header 8 bits	Hop Limit 8 bits
Source Address (128 bits)		
Destination Address (128 bits)		
Payload (Extension Headers + Data)		

IPv6 Base Header

Key Facts

- Fixed size = 40 Bytes
- Simpler than IPv4 header
- Faster processing by routers
- No checksum field
- No fragmentation fields

Advantage

Fixed-size headers reduce router processing overhead.

Field 1: Version

Version Field

- Length = 4 bits
- Identifies IP version.
- Value for IPv6:



Purpose

Allows routers and hosts to recognize IPv6 packets.

Traffic Class Field

- Length = 8 bits
- Replaces IPv4 Type of Service (ToS).
- Identifies packet priority.
- Supports Quality of Service (QoS).

Applications

- Voice over IP (VoIP)
- Video Conferencing
- Streaming Media

Flow Label

- Length = 20 bits
- Identifies packets belonging to the same flow.
- Enables routers to provide special handling.

Flow Example

Video streaming session between two hosts.

All packets in the session can carry the same flow label.

Traffic Class vs Flow Label

Traffic Class	Packet Priority
Flow Label	Packet Stream Identification

Traffic Class

Determines how important a packet is.

Flow Label

Groups packets belonging to the same communication flow.

Field 4: Payload Length

Payload Length Field

- Length = 16 bits (2 Bytes)
- Indicates size of payload only.
- Base header is not included.

$$\text{Payload Length} \leq 65535 \text{ Bytes}$$

Difference from IPv4

IPv4 contains:

- Header Length
- Total Length

IPv6 needs only Payload Length because header size is fixed.

Field 5: Next Header

Next Header Field

- Length = 8 bits
- Indicates what follows the base header.

Possible values:

- TCP
- UDP
- ICMPv6
- Extension Header



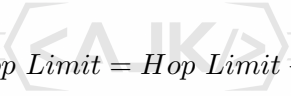
Similar To

Protocol Field in IPv4.

Field 6: Hop Limit

Hop Limit

- Length = 8 bits
- Replaces TTL field in IPv4.
- Decrement by every router.


$$Hop\ Limit = Hop\ Limit - 1$$

- If value becomes zero:
 - Packet is discarded.

Purpose

Prevents routing loops.

Fields 7 and 8: Source and Destination Address

Source Address

- Length = 128 bits
- Identifies original sender.

Destination Address

- Length = 128 bits
- Identifies final receiver.

Both addresses occupy:

$$128 + 128 = 256 \text{ bits}$$

Payload Contents

- 1 Zero or More Extension Headers
- 2 Upper Layer Data

Examples of Upper Layer Data:

- TCP Segment
- UDP Datagram
- ICMPv6 Message

Why Extension Headers?

IPv4 stores options inside the main header.

IPv6 moves options into separate extension headers.

Advantages:

- Faster routing
- Simpler base header
- Better scalability

IPv6 Payload Structure

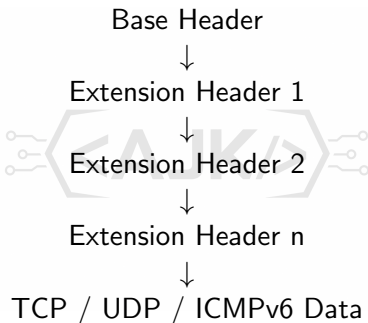
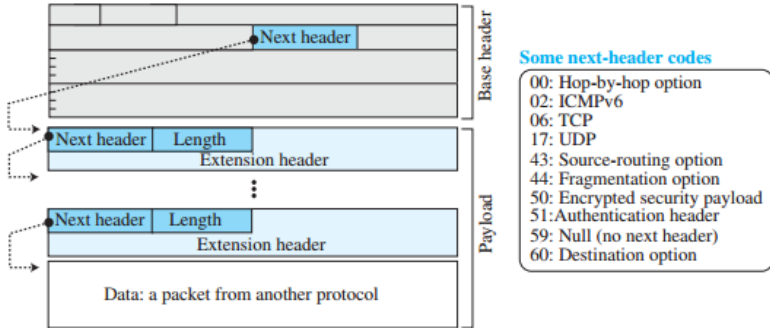


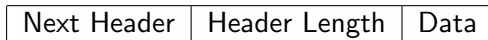
Figure 4.102 *Payload in an IPv6 datagram*



Extension Header Format

Each Extension Header Contains:

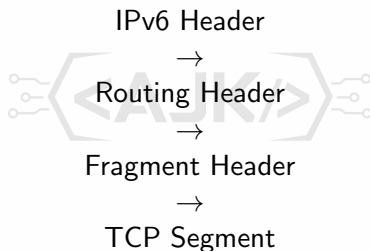
- Next Header Field
- Header Length Field
- Option Information



Role of Next Header

The Next Header field forms a chain.

Example:



Each Next Header value identifies the type of the next header.

Common IPv6 Extension Headers

- Hop-by-Hop Options Header
- Routing Header
- Fragment Header
- Destination Options Header
- Authentication Header (AH)
- Encapsulating Security Payload (ESP)

Advantages of IPv6 Packet Format

- Fixed Header Length
- Faster Router Processing
- Flexible Extension Mechanism
- Better QoS Support
- Improved Security
- Easier Future Expansion

IPv4 vs IPv6 Header Comparison

Feature	IPv4	IPv6
Header Length	Variable	Fixed (40 Bytes)
Address Size	32 bits	128 bits
Options	Inside Header	Extension Headers
TTL	TTL Field	Hop Limit
QoS Support	ToS	Traffic Class + Flow Label
Protocol Field	Protocol	Next Header

Summary

- IPv6 packet contains a 40-byte fixed header.
- Major fields:
 - Version
 - Traffic Class
 - Flow Label
 - Payload Length
 - Next Header
 - Hop Limit
 - Source Address
 - Destination Address
- Extension headers provide flexibility.
- IPv6 improves routing efficiency and scalability.

Flow and Priority in IPv6

- IPv4 was originally designed as a connectionless protocol.
- Modern applications require connection-oriented services.
- IPv6 introduces the **Flow Label** field.
- Flow Label enables routers to identify packets belonging to the same communication flow.
- Supports:
 - Real-time audio
 - Video streaming
 - Voice over IP (VoIP)
 - Multimedia applications

Concept of a Flow in IPv6

- All packets belonging to the same session carry the same flow label.
- Routers recognize and process them consistently.

Flow Label Table in a Router

Flow Label	Service Type	Next Hop
101	Video Stream	R2
102	Voice Call	R3
105	Online Class	R5

- Router maintains a Flow Label Table.
- Each entry defines:
 - Required service
 - Resource allocation
 - Forwarding decision

Benefits of Flow Label

- Faster packet forwarding.
- Reduced routing-table lookups.
- Better Quality of Service (QoS).
- Supports multimedia communication.
- Enables resource reservation.

Example

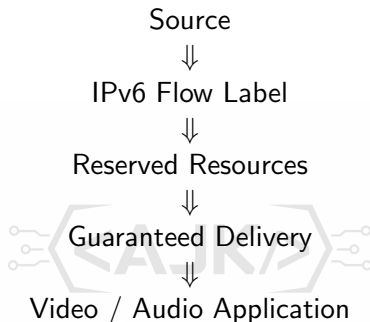
A live online lecture can reserve bandwidth and buffering resources for smooth video delivery.

Traffic Class and Flow Label

Traffic Class	Packet Priority
Flow Label	Packet Flow Identification

- Traffic Class determines priority.
- Flow Label identifies packets of the same communication stream.

Support for Real-Time Applications



Examples:

- Video Conferencing
- VoIP
- Live Streaming
- Online Gaming

IPv6 Fragmentation

Major Change from IPv4

Only the source host can fragment packets.

- Routers do NOT fragment packets.
- Reassembly occurs only at destination.
- Improves router performance.

IPv4 vs IPv6 Fragmentation

Feature	IPv4	IPv6
Fragmentation at Router	Yes	No
Fragmentation at Source	Yes	Yes
Reassembly	Destination	Destination
Processing Delay	Higher	Lower

IPv6 Fragmentation Process

- Router checks MTU.
- If packet is too large:
 - Packet is dropped.
 - ICMPv6 Packet-Too-Big message sent.
- Source retransmits fragmented packets.

Why Router Fragmentation is Removed

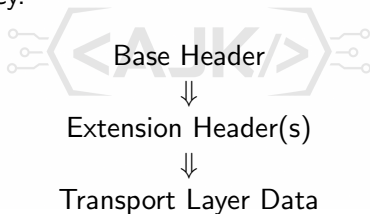
- Fragmentation requires additional computation.
- Router must:
 - Split packet
 - Update headers
 - Recalculate fields
- Increases delay.

IPv6 Goal

Keep routers simple and fast.

Extension Headers in IPv6

- IPv4 options are integrated into the header.
- IPv6 separates optional information into Extension Headers.
- Improves efficiency.



Structure of Extension Headers

Next Header	Header Length	Options/Data
8 bits	8 bits	Variable

Every extension header contains:

- Next Header field
- Length field
- Option-specific information

Important IPv6 Extension Headers

- 1 Hop-by-Hop Options Header
- 2 Routing Header
- 3 Fragment Header
- 4 Destination Options Header
- 5 Authentication Header (AH)
- 6 Encapsulating Security Payload (ESP)

Authentication Header (AH)

- Provides authentication.
- Verifies sender identity.
- Ensures data integrity.
- Prevents packet modification.

Security Benefit

Receiver can verify that the packet is from a legitimate sender.

Encapsulating Security Payload (ESP)

- Provides encryption.
- Ensures confidentiality.
- Protects sensitive information.

Examples:

- Banking Transactions
- VPN Communication
- Secure Video Calls

Advantages of IPv6 Flow Label and Extension Headers

- Faster routing decisions.
- Better Quality of Service.
- Improved multimedia support.
- Reduced router workload.
- Enhanced security.
- Flexible future expansion.

Summary

- Flow Label enables connection-oriented behavior in IPv6.
- Traffic Class provides packet prioritization.
- Only source hosts can fragment packets.
- Routers use ICMPv6 Packet-Too-Big messages.
- Extension Headers replace IPv4 options.
- AH and ESP provide authentication and security.

Why IPv6?

- IPv4 uses 32-bit addresses.
- Rapid Internet growth caused address depletion.
- IPv6 was developed to overcome IPv4 limitations.
- IPv6 uses 128-bit addresses.
- Provides a vastly larger address space.

$$2^{128} \approx 3.4 \times 10^{38}$$

Key Benefit

Address exhaustion is practically impossible.

IPv6 Address Representation

Length of IPv6 Address

- 128 bits
- 16 bytes
- 8 groups of 16 bits

Colon Hexadecimal Notation

FEF6:BA98:7654:3210:ADEF:BBFF:2922:FF00

- Preferred human-readable format.
- Eight sections separated by colons.

Binary Representation

1111111011110110 ... 1111111100000000

Colon Hex Representation

FEF6:BA98:7654:3210:ADEF:BBFF:2922:FF00

Observation

Binary notation is used internally by computers while colon hexadecimal notation is used by humans.

Table 4.7 *Prefixes for assigned IPv6 addresses*

<i>Block prefix</i>	<i>CIDR</i>	<i>Block assignment</i>	<i>Fraction</i>
0000 0000	0000::/8	Special addresses	1/256
001	2000::/3	Global unicast	1/8
1111 110	FC00::/7	Unique local unicast	1/128
1111 1110 10	FE80::/10	Link local addresses	1/1024
1111 1111	FF00::/8	Multicast addresses	1/256

Leading Zero Suppression

Leading zeros within a section can be removed.

Original	Abbreviated
0074	74
000F	F
0000	0
3210	3210

Rule

Only leading zeros may be removed.

Zero Compression

Consecutive groups of zeros can be replaced by ::

FDEC:0:0:0:0:BBFF:0:FFFF

becomes



FDEC::BBFF:0:FFFF

Important

Zero compression (::) can be used only once in an IPv6 address.

Mixed IPv6/IPv4 Notation

During IPv4-to-IPv6 transition, IPv4 addresses may be embedded.

::130.24.24.18

- Leftmost 96 bits are zeros.
- Rightmost 32 bits contain IPv4 address.

CIDR Notation in IPv6

IPv6 supports prefix notation.

FDEC::BBFF:0:FFFF/60

- Prefix length = 60 bits.
- Used for hierarchical routing.
- Similar to IPv4 CIDR notation.

IPv6 Address Space

$$2^{128}$$

possible addresses.

Compared with IPv4:


$$2^{128} = 2^{96} \times 2^{32}$$

- IPv6 address space is 2^{96} times larger than IPv4.
- Supports future Internet growth.

Types of IPv6 Addresses

IPv6 supports three address types.

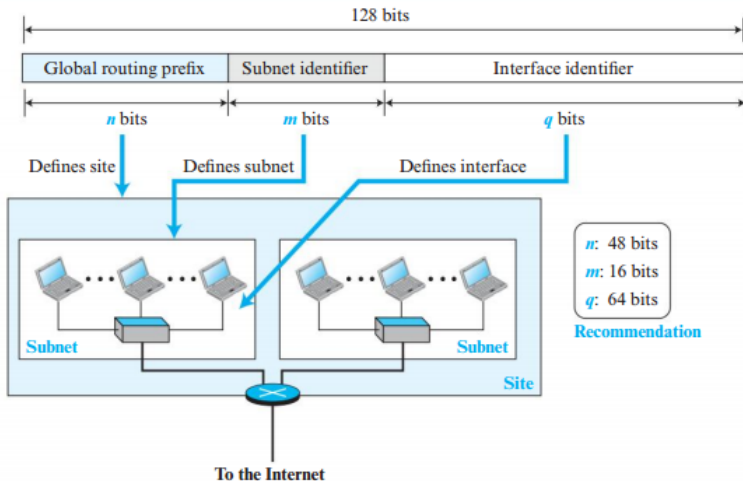
Type	Communication
Unicast	One-to-One
Anycast	One-to-Nearest
Multicast	One-to-Many

Unicast Address

- Identifies a single interface.
- Packet delivered to exactly one destination.
- Most common communication type.

Sender → Receiver

Figure 4.103 *Global unicast address*



Anycast Address

- Shared by multiple devices.
- Packet delivered to nearest or most reachable member.
- Commonly used for DNS and distributed services.

Client → Closest Server

Multicast Address

- Represents a group of receivers.
- Every member receives a copy.
- Efficient for video streaming and conferencing.

One Sender → Many Receivers

IPv6 Does Not Use Broadcast

- IPv4 supports broadcasting.
- IPv6 removes broadcast completely.
- Multicast is used instead.

Advantage

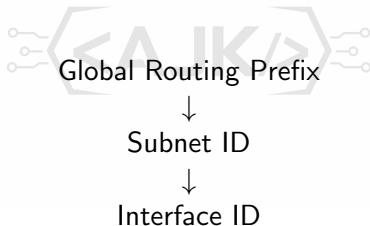
Reduces unnecessary network traffic.

Global Unicast Address

Global Unicast Prefix

2000::/3

Structure:



Global Unicast Address Structure

Global Routing Prefix	Subnet ID	Interface ID
48 bits	16 bits	64 bits

- Routing Prefix identifies organization.
- Subnet ID identifies subnet.
- Interface ID identifies host/interface.

Special IPv6 Addresses

Address	Purpose
::	Unspecified Address
::1	Loopback Address

Loopback and Unspecified Address

Unspecified Address

::

Used during initialization before obtaining an address.

Loopback Address

:::1

Used for testing local protocol stack.

IPv4-Compatible and IPv4-Mapped Addresses

Compatible Address

::IPv4-address

Mapped Address



::FFFF:IPv4-address

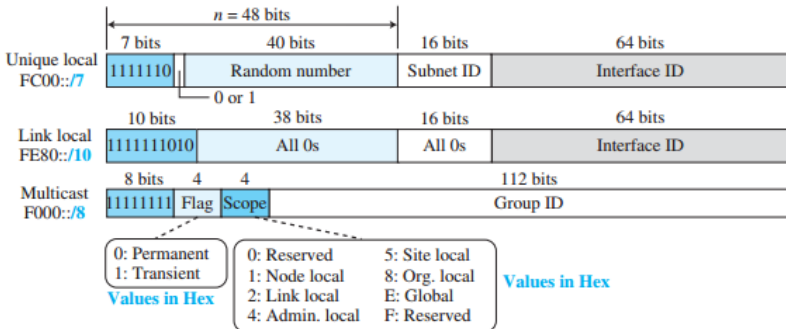
- Used during IPv4 to IPv6 transition.
- Allows interoperability.

Unique Local Unicast Addresses

Used for private networks.

- Similar to private IPv4 addresses.
- Not routable on the Internet.
- Intended for internal communication.

Figure 4.105 *Unique local unicast block*



Link-Local Addresses

- Used within a local network.
- Automatically configured.
- Not forwarded by routers.

Prefix:

FE80::/10

IPv6 Multicast Addresses

All multicast addresses begin with:

FF00::/8

- Used for group communication.
- Replaces broadcasting.
- Supports permanent and temporary groups.

IPv6 Address Allocation Overview

Block	Purpose
2000::/3	Global Unicast
FE80::/10	Link Local
FC00::/7	Unique Local
FF00::/8	Multicast
::1	Loopback
::	Unspecified

Summary

- IPv6 uses 128-bit addresses.
- Supports colon hexadecimal notation.
- Provides zero compression and CIDR notation.
- Address types:
 - Unicast
 - Anycast
 - Multicast
- No broadcast support.
- Large address space prevents address exhaustion.



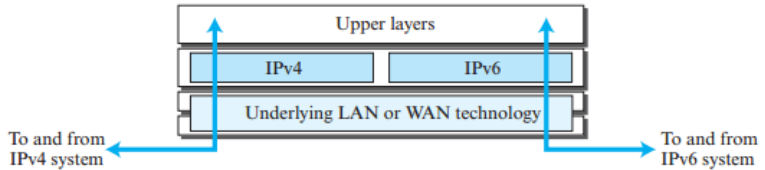
Transition from IPv4 to IPv6

- IPv4 and IPv6 cannot be replaced overnight.
- The Internet contains billions of hosts and routers.
- A sudden migration would disrupt communication.
- Therefore, transition must be gradual and smooth.
- IETF proposed three transition mechanisms:
 - Dual Stack
 - Tunneling
 - Header Translation

Dual Stack

- During the transition period, devices run both IPv4 and IPv6 protocol stacks simultaneously.
- A host can communicate using either IPv4 or IPv6 depending on the destination.
- DNS helps determine which protocol version should be used.
- If DNS returns an IPv4 address, the source sends an IPv4 packet.
- If DNS returns an IPv6 address, the source sends an IPv6 packet.
- Dual stack ensures backward compatibility while enabling IPv6 deployment.

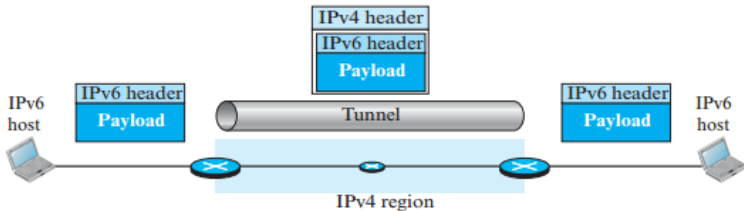
Figure 4.106 *Dual stack*



Tunneling

- Tunneling is used when two IPv6 networks communicate through an intermediate IPv4 network.
- IPv6 packets cannot travel directly through an IPv4-only region.
- Therefore, the IPv6 packet is encapsulated inside an IPv4 packet.
- At the tunnel entry point:
 - IPv6 packet becomes payload of an IPv4 packet.
- At the tunnel exit point:
 - IPv4 header is removed.
 - Original IPv6 packet is recovered.
- Protocol field value 41 indicates that the IPv4 packet carries an IPv6 packet.

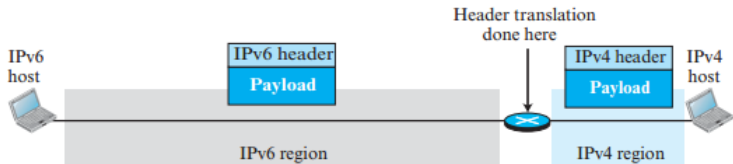
Figure 4.107 *Tunneling strategy*



Header Translation

- Header Translation is used when an IPv6 host communicates with an IPv4-only host.
- The destination device cannot understand IPv6 packets.
- Therefore, the IPv6 packet header must be converted into an IPv4 header.
- Unlike tunneling, the packet format itself changes.
- The translator converts:
 - IPv6 Source Address
 - IPv6 Destination Address
 - Header Fieldsinto corresponding IPv4 values.
- This enables communication between IPv6 and legacy IPv4 systems.

Figure 4.108 *Header translation strategy*



Comparison of IPv4 to IPv6 Transition Mechanisms

Mechanism	Purpose	Operation
Supports both protocols simultaneously	Device runs IPv4 and IPv6 stacks together	Dual Stack
IPv6 packet encapsulated inside IPv4 packet	Tunneling	Connect IPv6 networks through IPv4 infrastructure
Header Translation	Enable IPv6 and IPv4 communication	IPv6 header converted into IPv4 header

Goal: Achieve smooth migration from IPv4 to IPv6 without interrupting Internet services.

Key Points

- IPv6 was introduced to overcome IPv4 limitations.
- Transition to IPv6 is gradual and cannot occur instantly.
- Dual Stack allows devices to support both versions.
- Tunneling enables IPv6 communication through IPv4 networks.
- Header Translation allows interoperability between IPv4 and IPv6 hosts.
- These mechanisms ensure uninterrupted Internet communication during migration.

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- UDP and TCP

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ICMPv6: Internet Control Message Protocol Version 6

- ICMPv6 is the updated version of ICMP used in IPv6.
- It provides:
 - Error reporting
 - Diagnostic functions
 - Neighbor discovery
 - Multicast group management
- ICMPv6 combines the functionality of:
 - ICMPv4
 - ARP
 - IGMP
- It is more powerful and integrated than ICMPv4.

Figure 4.109 *Comparison of network layer in version 4 and version 6*



ICMPv6 Message Categories

ICMPv6 messages are divided into four major categories:

- ① Error Reporting Messages
- ② Informational Messages
- ③ Neighbor Discovery Messages
- ④ Group Membership Messages

Error Reporting Messages

Error-reporting messages notify the source when problems occur during packet delivery.

- Destination Unreachable
- Packet Too Big
- Time Exceeded
- Parameter Problem

Important Change:

- Source Quench message from ICMPv4 was removed.
- Packet Too Big message was introduced in IPv6.

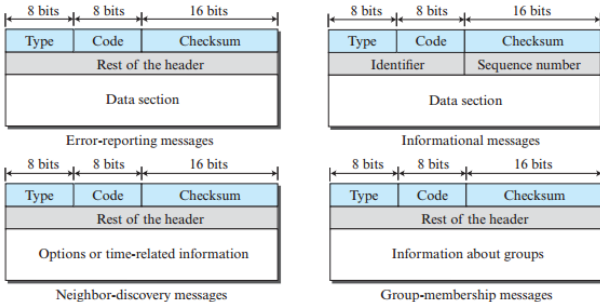
Reason:

- Routers are not allowed to fragment packets in IPv6.
- Oversized packets are dropped.
- Router informs source using Packet Too Big message.

Packet Too Big Message

- IPv6 routers cannot perform fragmentation.
- If packet size exceeds MTU:
 - Packet is discarded.
 - Router generates Packet Too Big message.
- Source host receives the error.
- Source retransmits packet using smaller size.

Figure 4.110 ICMPv6 messages



Type and code values

Error-reporting messages

- 01:** Destination unreachable (codes 0 to 6)
- 02:** Packet too big (code 0)
- 03:** Time exceeded (codes 0 and 1)
- 04:** Parameter problem (codes 0 to 2)

Neighbor-discovery messages

- 133 and 134:** Router solicitation and advertisement (code 0)
- 135 and 136:** Neighbor solicitation and advertisement (code 0)
- 137:** Redirection (codes 0 to 3)
- 141 and 142:** Inverse neighbor solicitation and advertisement (code 0)

Informational messages

- 128 and 129:** Echo request and reply (code 0)

Group-membership messages

- 130:** Membership query (code 0)
- 131:** Membership report

Note: See the book website for more explanation about messages.

Informational Messages

Informational messages provide diagnostic capabilities.

Supported Messages

- Echo Request
- Echo Reply

Applications:

- Connectivity testing
- Network troubleshooting
- Ping utility

Removed Messages

- Timestamp Request
- Timestamp Reply

These messages were rarely used and therefore removed in IPv6.

Neighbor Discovery Messages

Neighbor Discovery is a new feature of ICMPv6.

Functions include:

- Discover routers
- Discover neighboring hosts
- Resolve link-layer addresses
- Obtain IPv6 addresses
- Redirect traffic

Neighbor Discovery replaces:

- ARP
- Router Discovery
- Parts of DHCP functionality

Router Discovery Messages

Router Discovery helps hosts locate default routers.

Messages

- Router Solicitation (RS)
- Router Advertisement (RA)

Operation:

- 1 Host sends Router Solicitation.
- 2 Router responds with Router Advertisement.
- 3 Host learns:
 - Default router address
 - Network information
 - Configuration parameters

Address Resolution Messages

ICMPv6 replaces ARP using:

- Neighbor Solicitation (NS)
- Neighbor Advertisement (NA)

Purpose:

- Find MAC address from IPv6 address.
- Verify neighbor reachability.
- Maintain neighbor cache.

Benefits:

- Reduced network traffic.
- Improved efficiency.
- Better support for IPv6 networks.

Inverse Address Resolution

ICMPv6 also supports:

- Inverse Neighbor Solicitation
- Inverse Neighbor Advertisement

Purpose:

- Determine IPv6 address when MAC address is known.
- Useful in automatic configuration.

Redirection Messages

Redirection messages help optimize routing.

Operation:

- Router detects a better path.
- Router sends Redirect message.
- Host updates its routing information.
- Future packets follow a more efficient route.

Benefits:

- Reduced delay
- Better network performance
- Efficient packet forwarding

Group Membership Messages

Group Membership Messages support multicasting.

Messages:

- Membership Query
- Membership Report

Functions:

- Discover multicast group members.
- Manage multicast communication.
- Replace IGMP functionality of IPv4.

Applications:

- Video streaming
- Online conferencing
- Distance learning
- IPTV services

ICMPv4 vs ICMPv6

Feature	ICMPv4	ICMPv6
Error Reporting	Yes	Yes
ARP Support	Separate Protocol	Integrated into ICMPv6
IGMP Support	Separate Protocol	Integrated into ICMPv6
Neighbor Discovery	No	Yes
Router Discovery	Limited	Yes
Packet Too Big Message	No	Yes
Timestamp Messages	Yes	Removed
Fragmentation Support	Routers Can Fragment	Source Only Fragments
Multicast Management	IGMP Required	Built into ICMPv6

Observation: ICMPv6 combines the functionalities of ICMPv4, ARP, and IGMP while introducing Neighbor Discovery and improved error handling.

Key Points

- ICMPv6 combines ICMP, ARP, and IGMP functionalities.
- Supports error reporting and diagnostics.
- Introduces Neighbor Discovery mechanism.
- Handles router discovery and address resolution.
- Supports multicast group management.
- Packet Too Big message replaces router fragmentation.
- Essential protocol for efficient IPv6 operation.